

Vansh Jangir

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EXPERIENCE

Karya - Software Development Engineer - I | TypeScript, React, Postgres Jan 2026 - Present

- Built and deployed a full-stack video conferencing solution using Daily.co APIs and webhooks to enable (10,000+ hours) audio data collection via session recordings.
- Migrated audio processing pipeline from monolithic backend to Google Cloud Functions (GCP Lambda) to offload compute-heavy workloads, reducing backend load by 80%.

Karya - SWE Intern | TypeScript, React, Python, FastAPI, Redis, Postgres Feb 2025 - Dec 2025

- Developed a full-stack, Natural language-to-SQL web application powered by LLM APIs, improving report generation time by 50%, using FastAPI, SQLAlchemy, Celery, and a React-based conversational UI.
- Designed and implemented a query management system with persistent storage and result export, for 100+ users to ease collaborative data analysis.
- Re-architected backend services by trading off additional network and disk I/O to reduce CPU usage by 80% under production load.

ContractKen - SDE Intern | Python, Flask, Redis, React, TypeScript Aug 2024 - Nov 2024

- Built multiple legal document formatting tools that reduced document drafting and review time for 200+ active users.

PROJECTS

Xdb: Database and storage engine in Go | [github](#)

- A self-contained **ACID**-compliant database and storage engine in Go, supporting atomic transactions and data consistency, with 5,000+ write operations per second.
- Implemented **B+ tree** indexing and **Copy-on-Write**, improving write safety while maintaining sub-10ms read latency, and mmap for arbitrary disk access.
- Created a command-line interface (CLI) for database interaction, supporting basic SQL queries and transactional commands (begin, commit, rollback).

VanGo: An online Go game playing server | [website](#) [github](#)

- Developed a real-time multiplayer Go web application with user authentication (Google, guest), live gameplay against users, matchmaking and spectate mode.
- Architected a Go (Gin) + PostgreSQL + **WebSocket** backend with **Redis pub/sub**, maintaining sub-200ms move latency.

Go routines in C | [github](#)

- A Preemptive user-space scheduler in C inspired by Go's runtime model for x86-64 systems.
- Epoll-based event poller to handle network I/O blocked and timer blocked functions.

Load Balancer | [github](#)

- A **multi-threaded** layer-7 load balancer in Rust using low-level libc bindings for routing HTTP requests.
- Integrated an event-driven architecture to handle high connection concurrency with stable CPU usage.

EDUCATION

IIIT Naya Raipur

B.Tech. in Data Science and Artificial Intelligence, CGPA: 8.43/10

Raipur, Chhattisgarh

2022 – 2026

TECHNICAL SKILLS

Languages: C, C++, TypeScript, Python, Go, Rust, SQL.

Frameworks: ReactJS, ExpressJS, NextJS, Gin, Go-chi, Flask, FastAPI, HTML, CSS

Tools: GNU/Linux, Git, Docker, AWS, MongoDB, Make, CMake, Vim/Neovim, POSIX threads, Socket Programming.

Courses: Operating Systems, Distributed Systems, Data Structures, DBMS.